Note 11 Seasonal ARIMA Models

1 Seasonal ARIMA Models

Recall: An ARIMA(p, d, q) model is

$$\phi(B)(1-B)^d X_t = C + \theta(B) Z_t.$$

ARIMA can be used to model the **trend** in time series data. However, some series also have **seasonal effects** besides trend.

Let $\{X_t\}$ be a seasonal time series with period T, the most general model for $\{X_t\}$ has the form

$$\Phi(B^T)\phi(B)(1-B^T)^D(1-B)^d X_t = C + \Theta(B^T)\theta(B) Z_t.$$

This model is called the **multiplicative seasonal ARIMA model**, denoted as ARIMA$(p, d, q) \times (P, D, Q)_T$

2 Parameter Redundancy

3 Causality and Invertibility

Let $Y_t = (1 - B^T)^D (1 - B)^d X_t$, then $\{Y_t\}$ is causal (hence stationary) if

$\{Y_t\}$ is invertible if

Example 3.1. For the models below, determine p, d, q, P, D, Q and T. Also determine if the models are causal/invertible.

(a) $X_t - 1.4X_{t-1} - 1.2X_{t-2} = (1 - 0.8B^{12})(1 - 1.5B - B^2)Z_t$

(b) $(1 - 0.8B)(1 - B)^2 X_t = 5 + Z_t + 1.5Z_{t-1} - Z_{t-2}$

(c) $(1 - 2.6B - 1.2B^2)(1 - B^{12})(1 - B)X_t = Z_t - 0.35Z_{t-1}$

(d) $X_t + 0.3X_{t-1} - 0.1X_{t-2} = 5 + (1 + 0.75B^{12})Z_t$

(e) $(1 - 0.5B^4)(1 - 1.2B)(1 - B^4)X_t = 9 + Z_t - 1.4Z_{t-1} - 1.2Z_{t-2}$

4 Model Identification

Suppose a series $\{X_t\}$ (non-stationary, with trend and seasonal effect) is modeled by a causal and invertible ARIMA$(p, d, q) \times (P, D, Q)_T$ model, how to identify the order?

1. Find d, D and T from the plot and context of the series.

2. Once d, D and T are identified, take difference $Y_t = (1 - B^T)^D (1 - B)^d X_t$. $\{Y_t\}$ is a stationary series. According to the ACF and PACF of $\{Y_t\}$, find p, q and P, Q.

5 Estimation

MLE is the most recommended estimation method.

6 Diagnostics

Diagnostics are the same as ARMA/ARIMA models.

7 Forecast